

Task 3 Paper Review and Reproducibility Note

Fast High-Resolution Image Synthesis with Latent Adversarial Diffusion Distillation

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1. Problem and Motivation

Diffusion models produce high-quality images, but they are often slow at inference time because generation usually requires many sequential denoising steps. This creates a practical bottleneck for interactive image synthesis, editing, and deployment. The paper targets this speed-quality trade-off directly. Its core question is whether a strong multi-step text-to-image diffusion model can be distilled into a much faster generator without losing too much quality. This is a practically important problem because a method that preserves useful visual fidelity at very low step counts could make diffusion systems more usable in real-world settings.

2. Core Method

The paper proposes Latent Adversarial Diffusion Distillation (LADD). The key idea is to distill a stronger diffusion teacher into a fast student model while moving the adversarial supervision into a latent feature space rather than relying only on pixel-level matching. Compared with earlier adversarial diffusion distillation methods, LADD uses generative features from pretrained latent diffusion models. This design is intended to simplify optimization and improve visual quality, especially for high-resolution and multi-aspect-ratio synthesis. In broad terms, the method keeps the low-step generation goal of earlier distillation approaches but uses a more suitable representation for training supervision.

3. Data, Evaluation, and Main Findings

The paper studies high-resolution text-to-image generation and reports results on a distilled system derived from Stable Diffusion 3. The main reported outcome is that LADD enables SD3-Turbo, a model that can match strong contemporary text-to-image performance using only four unguided sampling steps. The paper also discusses scaling behavior and shows applications such as image editing and inpainting. The main contribution is therefore not only a new training objective, but also evidence that fast distilled diffusion can remain visually competitive under severe step reduction.

The evaluation matters because the paper is making a speed-quality claim rather than only a modeling claim. A method like LADD is only convincing if it can show that aggressive step reduction still preserves useful image quality at realistic resolutions. For that reason, the paper's evaluation combines qualitative visual comparisons with broader performance evidence, so readers can judge whether fast generation merely produces recognizable images or whether it remains competitive in fidelity and usability. This is also why my review focuses on step count as a key variable: the number of sampling steps is directly tied to inference speed, and it is the clearest practical axis along which the paper argues it improves text-to-image generation.

4. Two Strengths

First, the paper addresses a highly relevant systems-level limitation of diffusion models: inference cost. This makes the contribution practical rather than purely theoretical. Second, the method is conceptually well motivated. By using latent generative features instead of fixed pixel-space adversarial supervision, the paper proposes a training signal that is more naturally aligned with latent diffusion pipelines and high-resolution image synthesis.

5. Two Limitations and One Reproducibility Challenge

One limitation is that the full training setup is expensive and difficult to reproduce in a course environment. Recreating the method would likely require large models, significant compute, and access to the exact checkpoints or training configuration. A second limitation is that the paper's strongest claims are

supported by large-scale qualitative and benchmark evidence, which makes a small independent reproduction necessarily partial. The main reproducibility challenge is therefore not understanding the idea, but reproducing the original training and evaluation pipeline faithfully enough to verify the full reported gains.

What blocked full reproduction was not just implementation effort, but missing access to the paper's original training regime at the necessary scale. A faithful reproduction would require the exact teacher-student setup, large-scale training compute, matching pretrained checkpoints, and the same evaluation pipeline used to support the reported gains. In this course setting, I could test the general idea behind low-step generation, but I could not realistically retrain or benchmark the full system under comparable conditions. That is why the reproduction component in this submission is framed as a structured proxy experiment rather than a claim of end-to-end replication.

6. Small Reproducibility Component

Because full paper-scale reproduction is not feasible here, I include a compact proxy experiment that tests one central claim behind the paper: whether very small inference-step counts can still produce coherent images, and how image quality changes as the number of steps increases. The figure below compares outputs generated with 1, 2, and 4 steps using the same city-at-sunset prompt.

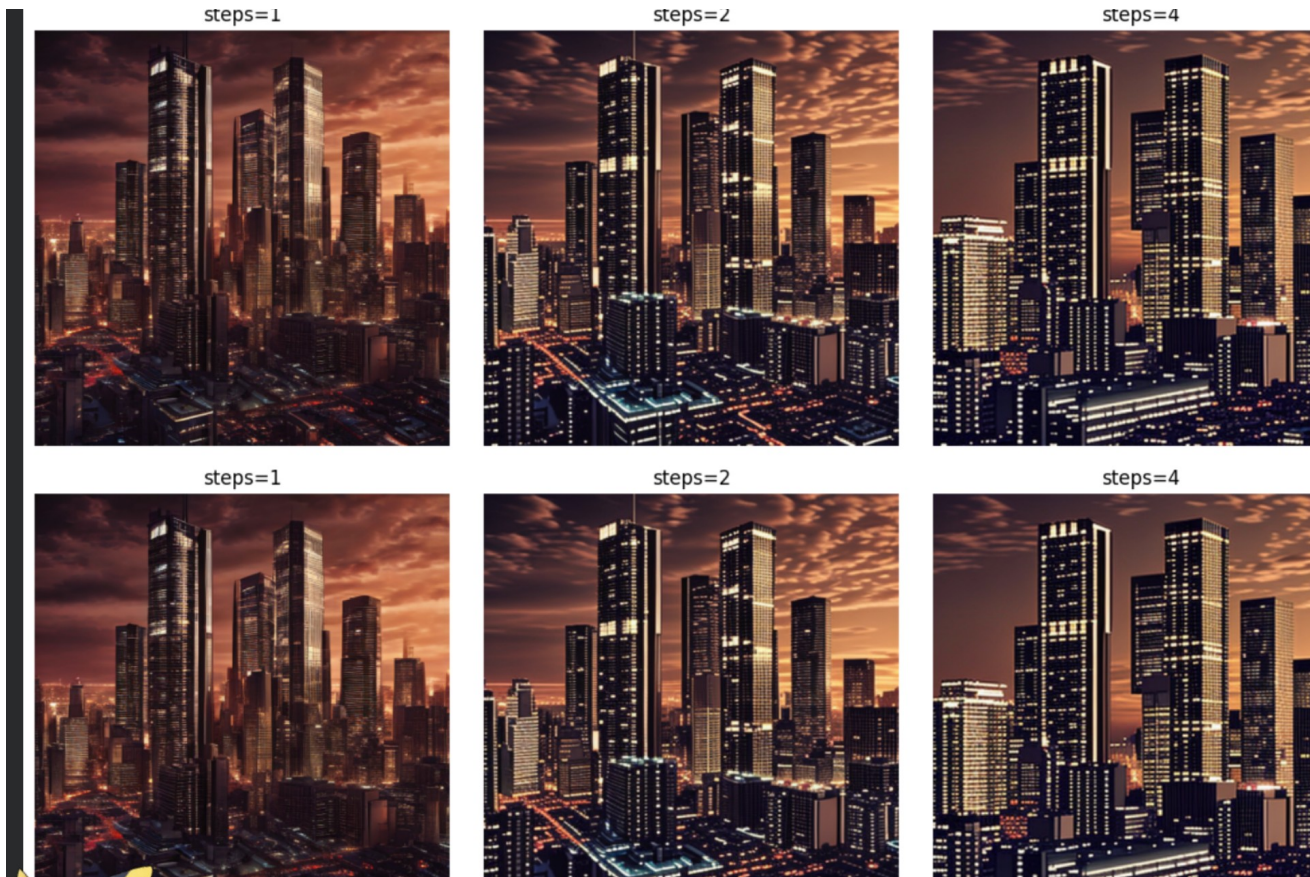


Figure 1. Proxy step-count comparison from the notebook: 1-step, 2-step, and 4-step outputs for the same prompt.

These images are included because a table of settings alone would not show whether the generated scene remains structurally recognizable under aggressive step compression. The 1-step image already captures the global concept of a lit city skyline at sunset, which is important because it shows that extreme speed-up can still preserve coarse semantic structure. At 2 steps, the buildings, windows, and lighting patterns become more regular, and the geometry of the skyline is visibly more stable. At 4 steps, the image is the most refined among the three: building edges are cleaner, window patterns are more consistent, and the overall output appears more polished. This supports the paper's broader motivation that fast distilled

diffusion can produce usable images at very low step counts, while also making the remaining quality trade-off visible.

7. Follow-up Experiment

A concrete next step would be to extend the qualitative step-count comparison with a small quantitative study. For example, I would compare runtime together with an image-quality proxy across 1, 2, 4, and possibly 8 steps, and then analyze where the diminishing-return region begins. That would strengthen the review because it would connect the visual evidence more directly to the speed-quality trade-off that motivates the paper.

References

Sauer, A., Boesel, F., Dockhorn, T., Blattmann, A., Esser, P., & Rombach, R. (2024). Fast High-Resolution Image Synthesis with Latent Adversarial Diffusion Distillation. [arXiv:2403.12015](https://arxiv.org/abs/2403.12015).